SWATANTRA DHARA

MS by Research in Artificial Intelligence and Data Science
Center For Brain Science and Applications,
School of Artificial Intelligence and Data Science,
Indian Institute of Technology Jodhpur

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Education

MS by Research in Artificial Intelligence and Data Science
(January 2024 – July 2026)
Center For Brain Science and Applications, School of Artificial Intelligence and Data Science, Indian Institute of Technology Jodhpur, Jodhpur, India

- CGPA: 7.95 (out of 10)
- Teaching Assistant Duties: Introduction to Cognitive Science (2025 Autumn), Machine Learning with Big Data (2025 Spring) Engineering Design (2024 Autumn)
- Non-Acad. Positions: Founder and head of The NeuroTech Society, Student Wellbeing Committee member.

MSc in Medical Biotechnology and Bioinformatics
(August 2021 – July 2023)
Center for Health Science And Technology, JIS Institute of Advanced Studies & Research, JIS University, Kolkata, India

- CGPA: 9.18 (out of 10), Rank: 1st, JIS Merit Scholarship, JIS Best Student Award

BSc Hons. Biotechnology
(August 2018 – July 2021)
Shoolini University, Solan, IN

- CGPA: 8.2 (out of 10)

High School Diploma, Sciences (Maths, Biology, Chemistry, Physics)
(2018)
West Bengal Council of Higher Secondary Education, Howrah Vivekananda Institution

- Score: 75%

Research / Project Experiences

Cognitive Brain Dynamics Lab (link)
Indian Institute of Technology Jodhpur and National Brain Research Center, India
Senior Project Associate (July 2026 – Nov 2026)
Supervisor: Dr. Dipanjan Roy
Project title: "**Spatio-temporal characterization of network dynamics across healthy ageing and diseases**"
Executions:

- Data preprocessing, statistical analysis and finding relations of functional dynamics and network structural parameters with healthy ageing across tasks, and clinical conditions (MCI, Alzheimer's), writing manuscripts.

Master's Student Researcher (Junior Research Fellow Equivalent)
(Jan 2024 – June 2026)
Supervisor: Dr. Dipanjan Roy
Thesis title: "**Characterizing Age-Associated Slowing Down of Speed of Transitions in Brain States**"
Executions:

- Time-series extraction, preprocessing, and denoising of fMRI data
- Fitting and interpretation of advanced statistical models on fMRI time series to understand functional network reconfiguration during movie watching, resting state and sensorimotor task, etc.

Medical Image Processing and Theranostics Laboratory (link)
Indian Institute of Technology Kharagpur, IN
Master's Thesis Student (Jan 2023 - June 2023)
Supervisor: Dr. Subhamoy Mandal
Thesis Title: "**Deep Learning Applications on Cutaneous Wound Healing From Optical Coherence Tomography Scans of Small Animals**"
Executions:

- Fitting of Deep Learning models (Autoencoders and Convolutional Neural Networks) on image data and statistical interpretation of results

- Manual segmentation of images

Genzel Lab (link), Donders Institute for Brain, Cognition and Behaviour,
Radboud University, Nijmegen, Netherlands
Neuroinformatics Intern
(November 2021 - June 2022)
Supervisor: Lisa K. A. Genzel
Project Title: "**Analyzing the effect of cannabidiol and anesthesia on sleep and memory formation"**
Executions:

- Wrote scripts in MATLAB for automatic assessment and analysis of sharp-wave-ripples(SWR) and non-rem sleep
- Fine-tuned the pre-existing SWR detection scripts

Kausik Basak's Lab, JIS Institute of Advanced Studies & Research, Kolkata, IN
Master's Minor Project
(March 2022 - June 2022)
Supervisor: Dr. Kausik Basak
Project title: **"Enzyme Class prediction from peptide statistics data with Artificial Neural Networks"**
Executions:

- Used ANN and Random forest to classify enzymes based on their chemical features
- Wrote Scripts to automate EMBOSS-Pepstats data collection pipeline

Yeast Biology Laboratory, Faculty of Applied Sciences & Biotechnology,
Shoolini University of Biotechnology and Management Sciences Dec 2020 - April 2021
Bachelor's Thesis Student
Project Title: **"Telomeres and Ageing"**
Supervisor: Prof. Anuradha Sourirajan
Executions:

- Major thesis writing on the structure of yeast telomere, telomere regulation, and how aging is connected with telomeres
- Learned and applied basic cell and molecular biology techniques on Yeast and E.Coli

Publications and Selected Conference Posters

"Deep Learning based Skin-layer Segmentation for Characterizing Cutaneous Wounds from Optical Coherence Tomography Images"

- **Authors:** Prashant Kumar, Swatantra Dhara, Ayan Gope, Jyotirmoy Chatterjee, Subhamoy Mandal
- **Journal:** 45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2023
- **Link:** https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10340321

"Age-associated slowing down of speed of transitions during naturalistic movie watching" (Poster)

- **Authors:** Swatantra Dhara, Suman Saha, Dipanjan Roy
- **Conference:** Annual Conference of Cognitive Science (ACCS12), BraiNDy 2026

Technical Skills

Neuroscience: fMRI Data Analysis, fMRI Preprocessing, Local Field Potential (LFP) Data Analysis, EEG Data Analysis, EEG Data Collection(OpenBCI), Basic Single Neuron Modelling, Spiking Neural Networks, EEG/EOG/EMG Data Collection with Arduino
Machine Learning/Deep Learning Models and Methods
Artificial Neural Networks, Segmentation, Convolutional Neural Network, Random Forests, K-Means Clustering, Classification, Regression, Deep Learning
Programming Languages: Python, MATLAB, R, Julia, C++
Softwares and Toolboxes: Nilearn, NiPy, CONN, SPM, fMRIprep, OpenViBE, Linux, Slurm Workload Manager, VS Code
Wet Lab Techniques: PCR, Restriction Endonuclease Assay, Transformation, Agarose Gel Electrophoresis, SDS-PAGE, Genomics and Plasmid DNA Isolation and Analysis, Antibiotic Susceptibility Testing, Cell Viability Assay, Cell Counting Assays, Cell Culturing Techniques

Professional Development

Volunteering and Organized Events

- ❖ Founder and Head of IITJ CBSA Neurotech Society (link)
  - ➢ Started a club on Neurotech at IITJ
  - ➢ Organized and lead two editions of SAIDE CBSA NeuroTech BCI hackathon for over 20, and 35 participants
  - ➢ Managed the planning and running of the weekly activities

- Member of BraiNDy 2026 Scientific Committee
- Neuromatch Academy and Neuromatch Conferences
  - Academy feedback analysis, Academy Issues Mediator(Discord), Social Events Handling, Outreach, Conference Planning and Outreach
- JISIASR: Founding and Head member of CHeST Student Club, successfully conducted a workshop on "Bioinformatics what and why?"
- Project Encephalon: Associate Chief Technology Officer
- Shoolini University: Project Envision (Chief Technology Officer)
- Howrah Vivekananda Institution: Member of Sahitya Sabha, Eco Club, Child Cabinet, Save Earth Campaign (Vice President)

Invited Talks

Invited Lecturer for International Youth Neuroscience Association's Neuroscience I & II Summer Programme:

- Memory and Learning (2026)
- Computational Neuroscience (2024 & 2025)
- Neuroimaging (2025)

Summer Schools and Workshops [Attended]

- Computational Approaches to Memory and Plasticity (CAMP) Workshop 2023, IISER Pune, supported by Simons Foundation (16 days)
- IEEE UpSkill EnMed 2022, IITKgp (2 days)
- Neuromatch Academy Computational Neuroscience 2021 (30 days)
  Project title: "Predicting image identities from neural activities"
- Neuromatch Academy Deep Learning 2021 (21 days)
  Project title: "Predicting visual fMRI responses to videos with a neuro-inspired network"
- 8th Baltic-Nordic Summer School on Neuroscience and Neuroinformatics 2021, HBP & EBRAINS (5 days)
- INCF Training Weeks 2021, INCF (5 days)
- International Conference on Data Science in Biology 2020, IITJ (5 days)

Massively Open Online Course Certificates

- Python Basics for Data Science, IBM, edX
- Statistics and R, HarvardX, edX
- Probability and Statistics: To p or not to p?, University of London, Coursera
- Writing in the Sciences, Stanford University, Coursera
- Principles of Biochemistry, HarvardX, edX
- Molecular Biology – Part 2: Transcription and Transposition, MITx, edX
- Epigenetics Control of Gene Expression, University of Melbourne, Coursera
- Introduction to Genomic Technologies, Johns Hopkins University, Coursera
- Fundamentals of Immunology: Innate Immunity and B-Cell Function, Rice University, Coursera

Exams

Qualified National Exams

- GATE-Biotechnology 2022, All India Rank: 486, top 3%
- IIT-JAM Biotechnology 2021, All India Rank: 540, top 7%
- TIFR-JGEEBILS 2021

Hobbies

Cycling, Badminton, Playing the Harmonica & the Keyboard, Running, Gardening, Reading Books, Cooking, Photography, Traveling, Meditating, and Finding Patterns in the Universe

Other Details

Languages: English (Fluent), Bengali (Native), Hindi (Fluent), German (Basics).